Big data processing and analytics

01DSHYG, 01DSHWQ, 01DSHWR

A.A. 2026/27

Course Language Inglese

Degree programme(s)
Master of science-level of the Bologna process in Ingegneria Informatica (Computer Engineering) - Torino
Master of science-level of the Bologna process in Cybersecurity Engineering - Torino
Master of science-level of the Bologna process in Cybersecurity Engineering - Torino

Course structure

Teaching	Hours

Lecturers

Teacher	Status	SSD	h.Les	h.Ex	h.Lab	h.Tut	Years teaching

Co-lectures Espandi

Context

SSD	CFU	Activities	Area context
ING-INF/05	6	B - Caratterizzanti	Ingegneria informatica

Anno accademico di inizio validità **2025/26**

Course description

In the big data era, novel data management and data analytic frameworks are needed. Specifically, to manage and fruitfully exploit the big amount of available heterogeneous data, novel data models, programming paradigms, data processing systems, and frameworks have been proposed.
The course addresses the challenges arising in the Big Data era. Specifically, the course will cover how to store, retrieve, and analyze big data to extract useful knowledge and hints. The course covers not only data models and data analytics aspects but also novel programming paradigms (i.e., MapReduce and its extensions) and discusses how they can be exploited to support big data engineers and scientists to manage and extract insights from data.

Expected Learning Outcomes

The course aims at providing:
• Knowledge of the main problems and opportunities arising in the big data context and technological characteristics of the infrastructures and distributed frameworks used to deal with big data (e.g., Hadoop and Spark).
• Ability to write distributed programs to process and analyze data by means of novel programming paradigms: MapReduce and Spark-based programming paradigms
• Knowledge of the (relational and non-relational) databases systems that are used to store big data

Pre-requirements

Python language, basic knowledge of Java language, and basic knowledge of traditional database concepts (relational model and SQL language).

Course topics

Lectures (45 hours)

Assessment and grading criteria

Exam: Computer-based written test in class using POLITO platform;

The exam aims at assessing (i) the ability of the students to write distributed programs to process and analyze big data by means of novel programming paradigms and frameworks (the MapReduce programming paradigm and the Spark-based programming paradigm) and (ii) the knowledge of the students of the main issues related to the big data topic and the technological infrastructures and distributed systems that are used to deal with big data.

The exam consists of a written onsite PC-based test (with the Exam platform) that lasts 1.5 hours.

Specifically, the written test (max 31 points) is composed of two parts:
- 2 programming exercises (structured as open questions) based on MapReduce- and Spark-based programming (27 points)
- 2 multiple-choice questions on all the topics addressed during the course (4 points).

The programming exercises aim at evaluating the ability of the students to write distributed programs to analyze big data through the novel programming paradigms introduced in the course.
The multiple-choice questions are used to evaluate the knowledge of the theoretical concepts of the course and the knowledge of the characteristics of the main technological infrastructures and distributed systems (Hadoop and Spark), including scalable relational and non-relational database systems, that are used to deal with big data.

The evaluation of the programming exercises is based on the correctness and efficiency of the proposed solutions.
The multiple-choice questions are evaluated on the correctness of the answer.

The exam is open book.